Stack and Queue

1) Which principle is used in the implementation of a stack? (1 Mark)

A) FIFO
B) LIFO
C) FCFS
D) Random Access

2) Which operation removes the topmost element from a stack? (1 Mark)

A) Push
B) Insert
C) Pop
D) Delete

3) Explain the concept of a circular queue with a neat, labelled diagram and describe how it differs from a linear queue (any one difference). (5 Mark)

4) Explain the working of a stack and describe its basic operations with a suitable example (5 Mark)

5) Explain the working of a queue and describe its basic operations with a suitable example. (5 Mark)

6) Write a C program to implement Stack using an array with the following operations: PUSH, POP, DISPLAY Evaluate your implementation by handling both Stack Overflow and Stack Underflow.

7) Write a C program to implement a Queue using an array with: INSERT, DELETE, DISPLAY Evaluate the implementation after multiple insertions and deletions. (5 Mark)

8) Using a stack, apply the procedure for converting the following infix expression into postfix notation**:** (5 Mark each)

a) A + B × (C − D)
b) (A − B) / (C + D) × E
c) A + B × (C − D)
d) A × (B + C) − D / E

9) Analyze the following postfix expression using a stack. Show the stack contents after each operation and determine the final result. (5 Mark each)

a) 8 4 2 + * 6 –
b) 7 3 2 * + 5 –
c) 9 5 2 + * 4 –
d) 9 5 2 + * 4 –